

Hamdi Barkous

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PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in financial AI systems, deep learning, and agentic frameworks. Experienced in building production-ready trading models, time series forecasting, and full-stack AI applications.

EXPERIENCE

Made With Chat - *Full-Stack / AI Engineer*

July 2025 – present

- Building an AI co-founder SaaS letting non-technical founders ship product changes via chat, voice, and async delegation.
- Architected a dual-brain agent system: a LiveKit voice agent (GPT-4.5) delegating complex coding tasks to a LangGraph deep agent (Claude) via a shared memory layer.
- Designed session-durable agent memory in a Daytona sandbox, eliminating conversation summarization and preserving cross-session context.
- Built a real-time SSE progress pipeline (LangGraph → Zustand) with human-in-the-loop interrupt/resume via the Command pattern.
- Shipped multi-channel surfaces (web, voice, WhatsApp, Telegram) on NestJS 11 + Next.js 16, with Langfuse observability and Stripe billing.

FinGenesis - *ML Engineer*

2024 – 2026

- Engineered DL&ML models to predict price movements across 500+ financial symbols, achieving a 2+ profit factor in live trading.
- Built an agentic sentiment analysis framework aggregating 10+ data sources per ticker to generate real-time trading signals.
- Architected rule-based and LLM-powered trading strategies with built-in explainability for risk management decisions.
- Created a backtesting engine validated against 5+ years of historical market data, enabling rapid strategy iteration.
- Streamlined data pipeline architecture, cutting processing time by 60%.

PROJECTS

Manim Agent - *Agentic Pipeline for Educational Videos* – 

Active

- Built a code-based multi-agent system for automated YouTube video generation, orchestrating planner, coder, reviewer, and critic agents to produce fully rendered Manim animations from a single prompt.
- Designed an end-to-end voiceover pipeline integrated with ElevenLabs TTS, synchronizing generated narration with scene timing for seamless audio-visual output.
- Powers the “Frame Zero” YouTube channel producing the *From Zero to Intelligence* AI / Math series.

IntoTeX - *AI-Powered PDF to LaTeX Converter* – 

Live Production

- Built a full-stack SaaS platform that converts PDF documents to LaTeX source code using a LangGraph-based AI agent with vision capabilities
- Engineered a PDF extraction engine capable of parsing complex document structures including formatted text, raster images, and embedded vector graphics
- Integrated a credit-based subscription system with Stripe payment processing

PUBLICATIONS

Barkous, H., Amayri, M., & Bouguila, N. “**A Comprehensive Analysis of a Hybrid Deep Learning Model for Midterm Electric Load Forecasting**” 21st IEEE International Conference on Smart Cities, Australia, Dec 13, 2023

EDUCATION

École Polytechnique de Tunisie – *ABET Accredited*

2020 – 2023

Multidisciplinary Engineering - Minor: Signals & systems

Relevant Courses:

- **Mathematics:** Optimization, Detection and Estimation, Probability & Statistics, Stochastic Processes
- **Computer Science:** Artificial Intelligence, Data Science, Data Analysis, Algorithms and Data Structures.
- **Signals and Systems:** Signal Processing, Digital Communication, Control Theory, Information Theory

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, SQL
- **ML/AI Frameworks:** LangChain, LangGraph, PyTorch, TensorFlow, OpenCV
- **API Development Stack:** FastAPI, NestJS, Next.js, PostgreSQL, Redis, Supabase
- **ML Platforms & Tools:** Hugging Face, MLflow, Langfuse, LiveKit
- **DevOps & Cloud:** AWS, Vercel, Docker, Git
- **Developer Tools:** Cursor, MCP